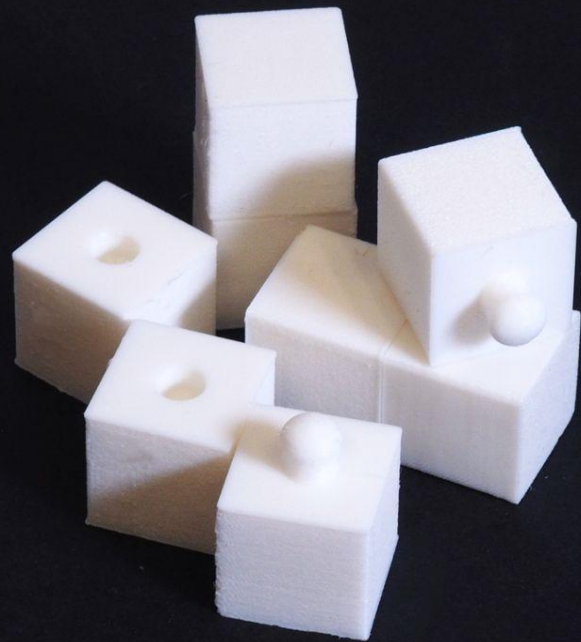




MOGO: Modular Soft Robotic Systems and Connection Techniques for Reconfigurable Assemblies

CMU 24-673 Final Presentation

Yujia Liu, Rohan Singh, Hongfei Liu, Brady Stump

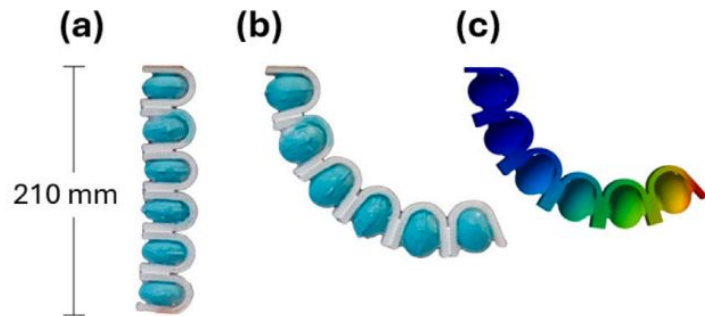
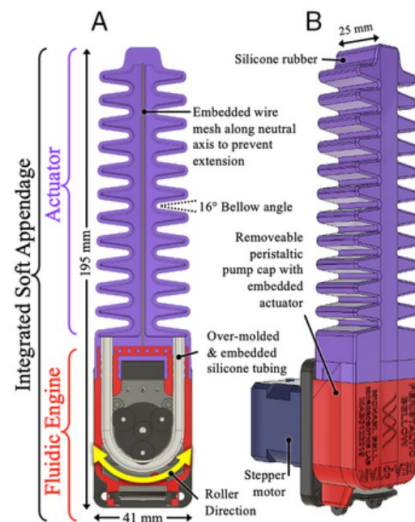
Short Recap

The Problem:

- Traditional robots are rigid, expensive, and unsafe around humans, soft robotics offers safe human interaction and adaptability
- But soft robots are difficult to design and fabricate
- Each application requires custom engineering from scratch

Our Vision:

- What if building soft robots was as intuitive as LEGOs?
- Enable rapid prototyping through modular, reconfigurable building blocks
- Make soft robotics accessible to researchers, educators, and makers



Our Journey

Where We Started:

Explored cable-driven actuation systems -
Investigated cable routing through flexible
modules

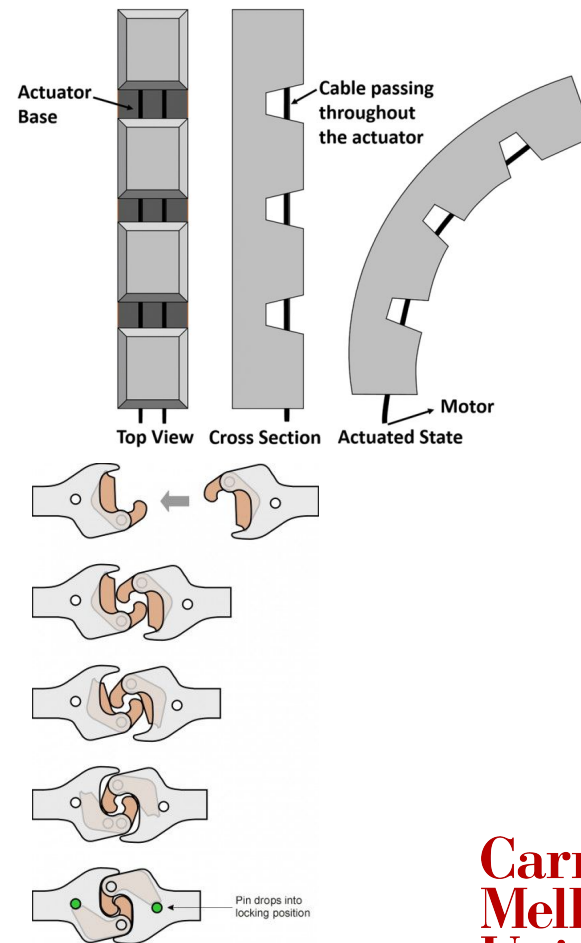
Focused on: "How do we make the actuation
work?"

The Pivot:

- Realized actuation wasn't the bottleneck
- The real challenge: How do modules connect?

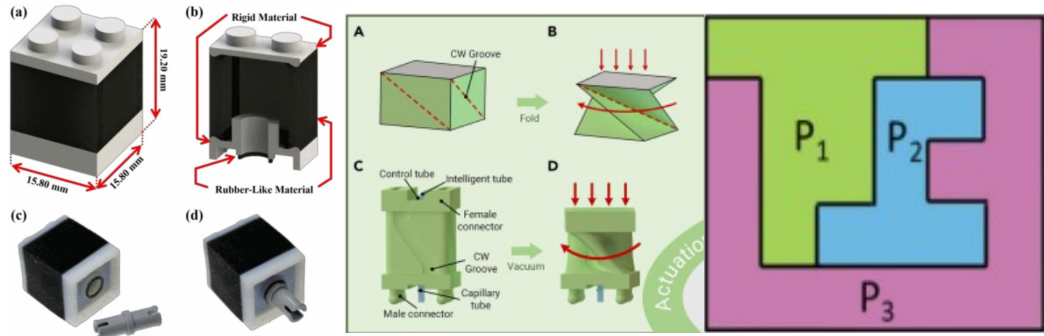
Key Question:

"What makes LEGOs successful?" → Easy +
Robust connections that anyone can use



Literature Comparison

Comparison to Closest Examples in Literature



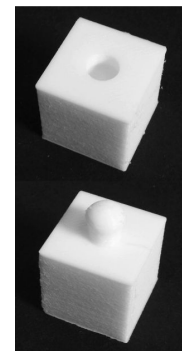
Approach	Key Features	Limitations
Soft LEGO [1]	Tango/Vero hybrid, LEGO-compatible	Rigid-soft interface limits compliance
Soft Origami LEGO [2]	E630/E650 elastomer, vacuum-actuated	Structure allows only twisting motion
Computationally Designed Interlocks [3]	Agilus30, computational design	Too complex for hands-on assembly

Literature Gap: The Connector Compliance Paradox

- All existing systems have **CCR (Connector Compliance Ratio) $\gg 100$**
- Rigid connectors in soft systems cause stress concentrations

MOGO's Position

- Single-piece, all-soft
- Reliable connection
- Allow bending & twisting
- Target CCR $\sim 1-10$



Experimental Methods

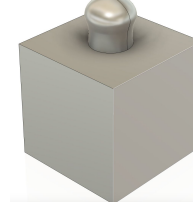
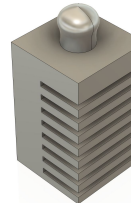
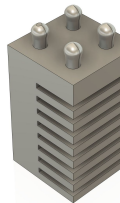
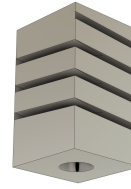
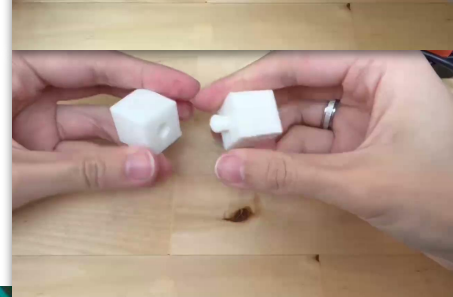
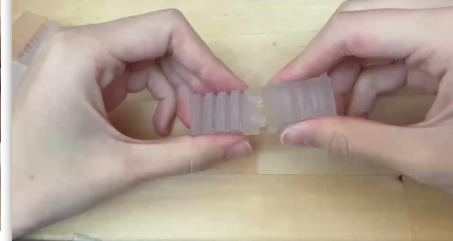
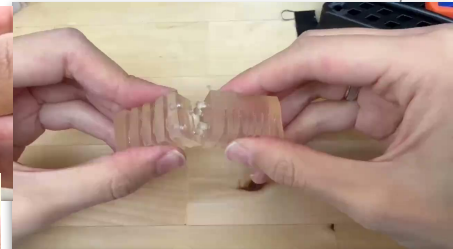
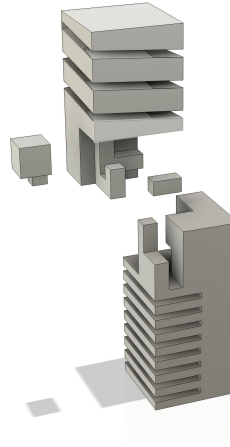
Equipments and Materials:

- Formlabs SLA 3D Printer & cleaning station
 - Formlabs Flexible Resin 80A
 - Isopropyl alcohol (IPA) for post-processing
 - UV curing light
- Anycubic Kobra 3
 - TPU (Thermoplastic Polyurethane) 30% infill

Purpose:

Simplify to focus only on the connection

Try different infill densities, interlocking structures, different parameters



Simulation for Designs/Testing

Comprehensive structural analysis of a 3D-printed Flexible Resin 80A and TPU blocks with interlocking mechanism under three loading conditions: torsion, bending, and tension.

"Hand tight" torque: $\sim 4\text{--}5\text{ Nm}$ (average adult)

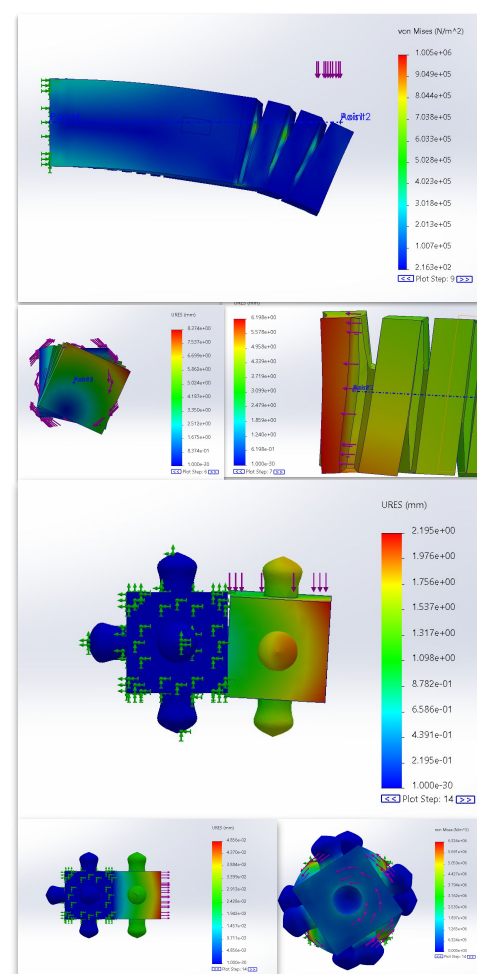
Source: NASA Anthropometric Source Book, citing Asmussen & Heeboll-Nielsen (1961), Backlund & Nordgren (1968), Laubach (1976)

Keyboard key press force: $\sim 0.50\text{ N}$ (50.6 grams)

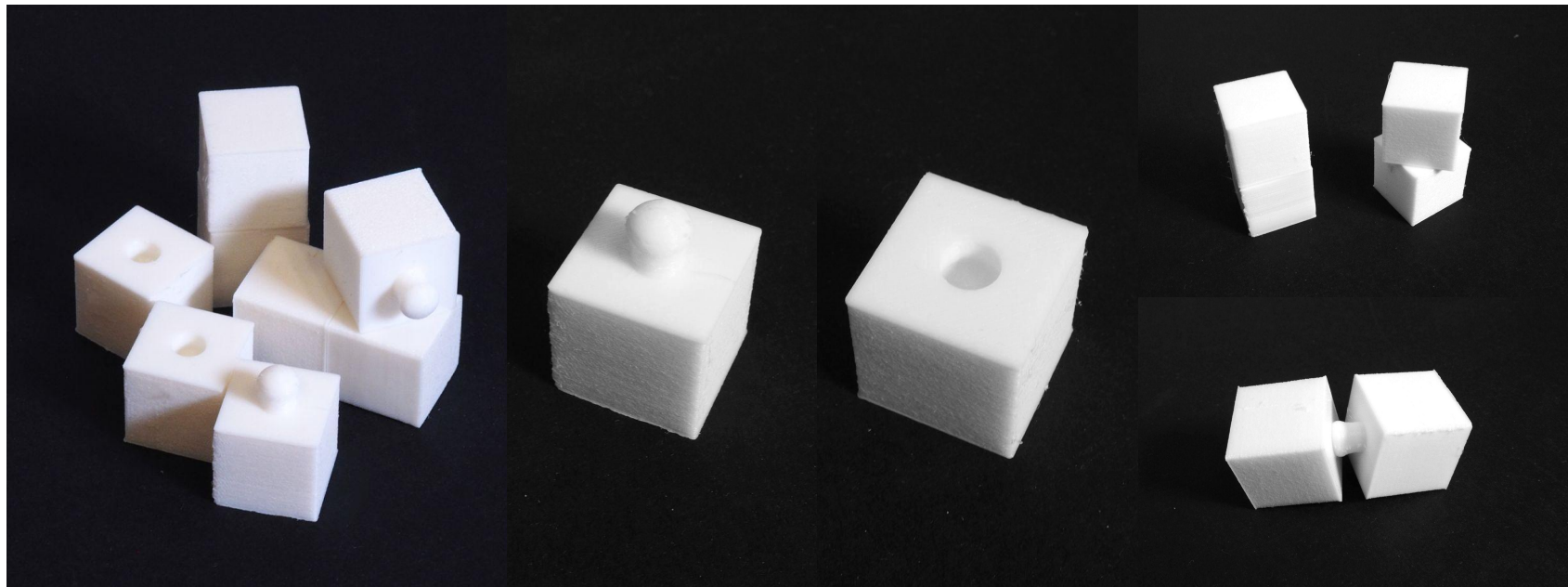
Source: RESNA Conference Proceedings (2004) - Study on relaxed finger weight for keyboard activation

Peak typing force: $\sim 2.54\text{ N}$ (263.1 grams)

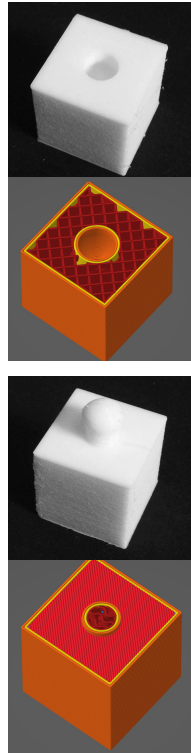
Source: University of Michigan study on typing dynamics



Final Version



Force Tests and Key Results



1. **Socket Module:**
Infill density: 20%-40%
Infill pattern: Rectilinear/Grid

Design rationale:

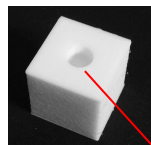
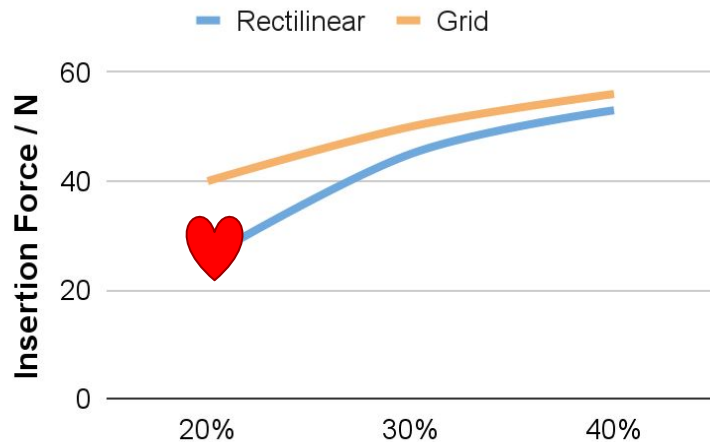
- Lower density + straight-line patterns increase compliance,
- Allowing the ball to **insert smoothly**.

2. **Ball Module:**
Infill density: 30%-60%
Infill pattern: Gyroid

Design rationale:

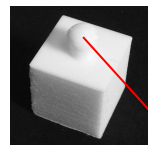
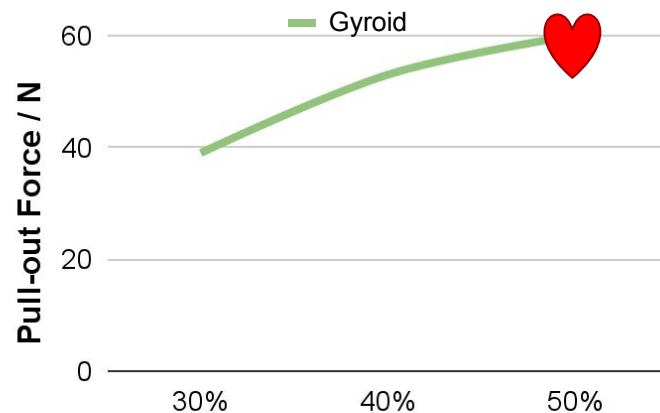
- Higher-density + gyroid pattern distributes stress evenly,
- Making the ball stay **stable during pulling**.

Figures of Merit



Infill Density of Socket Module

Soft edge -> easy to insert



Infill Density of Ball Module

Solid body -> stable to hold the shape

Research Contribution

Soft Ball-and-Socket Design: Made a compliant connector that enables secure, repeatable, and reconfigurable attachment.

Softness: Achieved controlled flexibility by 3D-printing the modules in TPU (Thermoplastic Polyurethane) with around 30% infill and different infill pattern, providing the deformation needed for secure connection.

Passive Alignment: Rounded head and tapered cavity provide geometric guidance for easier and reliable connections.

Compliance Enhances Robustness: The soft socket walls increase contact area, distribute stress, handle small misalignments, and create a vacuum suction during pulling.

The logo for Carnegie Mellon University is positioned on the left side of the slide. It features a dark blue background with a complex pattern of intersecting diagonal lines in red, green, and yellow. The text "Carnegie Mellon University" is written in a white, serif font, stacked in three lines.

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Thank you!
Questions?

Citations

1. Zhang, H., et al. (2020). "Modular Soft Robotics: Modular Units, Connection Mechanisms, and Applications." *Advanced Intelligent Systems*, 2(1), 1900166.
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5. ScienceDirect (2024). "Modelling of Modular Soft Robots: From a Single to Multiple Building Blocks." *Robotics and Autonomous Systems*.
6. Nature Materials (2025). "Scalable functionalized shape memory alloy fiber with synergistic effect for robotic hand and microrobot." *npj Flexible Electronics*.
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